



File Processing

- File is a named location on the system storage which records data for later access.
- It is used to store data permanently for future use to perform various operations like opening, reading, writing etc.
- Types of File:
 - Text File
 - Binary File

- A text file is simply a file which stores sequences of characters using an encoding like utf-8, latin1 etc. whereas in case of binary file, data is stored in the same format as in computer memory.
- Example text files are python source code, HTML file, text file etc. whereas executable files, images, audio etc are examples of binary files.

File Opening in Various Modes

Access Mode	Description
r	Open a file for reading (default mode). File must exist.
w	Open a file for writing. Creates a new file or truncates (overwrites) if it exists.
x	Open a file for exclusive creation. Fails if the file already exists.
a	Open a file for appending. Creates a new file if it doesn't exist.
b	Binary mode. Used with other modes for binary files. E.g., "rb" for reading binary.
+	Open a file for both reading and writing. Used with other modes. E.g., "r+".

Opening and Closing a file:



- The `open()` function is used to open a file. It returns a file object.

`file = open('filename.txt', mode)`

- **filename.txt**: The name of the file you want to open.
- **mode**: Specifies the purpose (reading, writing, appending, etc.).

Common modes:

- `'r'`: Read (default mode, file must exist).
- `'w'`: Write (creates a new file if it doesn't exist, truncates if it does).
- `'a'`: Append (creates a new file if it doesn't exist).
- `'r+'`: Read and write.

Reading from a file

- `read()`: Reads the entire content of the file.
- `readline()`: Reads a single line from the file.
- `readlines()`: Reads all the lines and returns a list.

```
file = open('filename.txt', 'r')
content = file.read()
print(content)
file.close()
```

```
file = open('filename.txt', 'r')
line = file.readline()
print(line)
file.close()
```

```
file = open('filename.txt', 'r')
lines = file.readlines()
print(lines)
file.close()
```

Writing to a file:

- `write()`: Writes data to the file.
- `writelines()`: Writes a list of strings to the file.

```
file = open('filename.txt', 'w')  
file.write('Hello, World!')  
file.close()
```

```
file = open('filename.txt', 'w')  
file.writelines(['Line 1\n', 'Line 2\n'])  
file.close()
```

Checking if a file exists

➤ You can use the os module to check if a file exists.

```
import os  
if os.path.exists('filename.txt'):  
    print('File exists')
```


Delete Existing File

➤ To remove a file ,you can use the remove() from os module.

```
import os
if os.path.exists('filename.txt'):
    os.remove('filename.txt')
else:
    print("file not exist")
```

- The seek() method in Python is used to move the file pointer to a specific position in a file. This allows you to control where in the file reading or writing starts. It's useful when you want to read/write at a specific point in a file, not necessarily from the beginning.

Working with binary file



Thank You

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